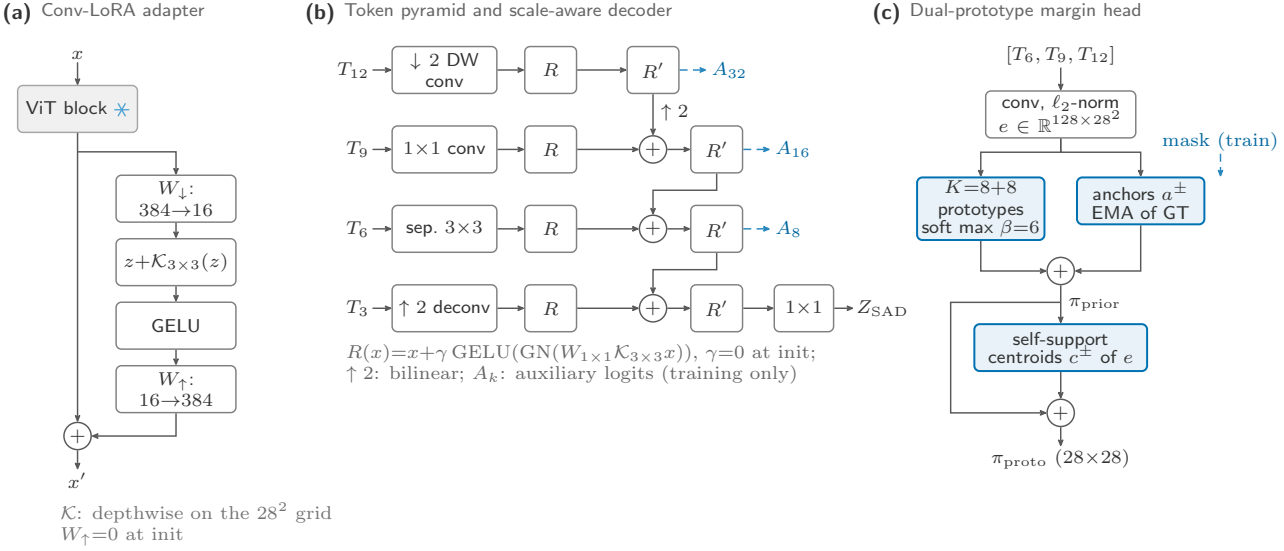

(a) Conv-LoRA adapter
x
ViT block
$W_{\downarrow}$: 384→16
$z+\mathcal{K}_{3\times3}(z)$
GELU
$W_{\uparrow}$: 16→384
$x'$
$\mathcal{K}$: depthwise on the $28^2$ grid
$W_{\uparrow}$=0 at init
(b) Token pyramid and scale-aware decoder
$T_{12}$
↓ 2 DW conv
$R$
$R'$
$A_{32}$
↑ 2
$T_9$
1×1 conv
$A_{16}$
$T_6$
sep. 3×3
$A_8$
$T_3$
↑ 2 deconv
1×1
$Z_{\mathrm{SAD}}$
$R(x)=x+\gamma\,\mathrm{GELU}(\mathrm{GN}(W_{1\times1}\mathcal{K}_{3\times3}x))$, $\gamma$=0 at init;
↑ 2: bilinear; $A_k$: auxiliary logits (training only)
(c) Dual-prototype margin head
$[T_6, T_9, T_{12}]$
conv, $\ell_2$-norm
$e \in \mathbb{R}^{128\times28^2}$
mask (train)
$K$=8+8 prototypes soft max $\beta$=6
anchors $a^{\pm}$ EMA of GT
$\pi_{\mathrm{prior}}$
self-support centroids $c^{\pm}$ of $e$
$\pi_{\mathrm{proto}}$ (28×28)